

Anonymisation Beyond PII: LLM-Adversarial Text Anonymisation via Auxiliary Information Linkage

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Abstract

Text anonymisation is dominated by methods that use Personally Identifiable Information (PII) as the unit of protection. Recent work has improved on this by using Large Language Models (LLMs) to extend protection from PII in its surface form to PII that can be inferred. However, text can contain no PII, either surface-level or inferable, and still allow re-identification of an individual. Such documents remain personal data under data protection law, which focuses upon outcome rather than the removal of enumerated identifiers. Yet, current state-of-the-art anonymisation methods would leave these documents unprotected.

This thesis replaces the adversary central to current LLM-adversarial anonymisation methods. Rather than inferring personal attributes, an ensemble of adversaries each hold a subset of auxiliary information or background knowledge and attempts to link it to the anonymised document. We propose ALTA, which uses this linkage based adversary, and ALTA-FD that extends it with a fine-grained rewriting process: a document is decomposed into a list of facts, each is transformed by generalisation or perturbation, and the transformed facts are reconstructed into a new document. Both methods are task-agnostic and require no downstream labels during anonymisation.

On DB-BIO, semantic leakages falls from 43.73 to 6.41 against the state-of-the-art, a reduction of over 85%, while distributional plausibility improves (MAUVE 0.587 to 0.963) and classification accuracy remains within 1.26 percentage points of a baseline conditioned on the ground-truth label. On MEDQA, a dataset of medical board examination questions, our method improves across all privacy and utility metrics from the baseline, notably improving question accuracy from 66.8% to 72.4% despite introducing perturbed clinical details. ALTA-FD improves privacy and task-specific utility over ALTA on both datasets, indicating that perturbation applied at the level of individual facts can recover information destroyed by generalisation.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Joseph Dunne)

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AI was used to discuss ideas and key themes in accordance with the Schools “AI-Assisted Editing” rule, that is applied to dissertations.

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Chapter 1

Introduction

The proliferation of LLMs has allowed inexpensive and automated processing of vast amounts of unstructured data such as text. This has raised concerns for privacy, where LLMs could be applied at scale to automatically re-identify individuals from anonymised datasets or pseudonymous websites or forums [16][7]. Privacy legislation like GDPR calls for data holders to account for future technological developments [2], but existing anonymisation methods are falling behind.

Current industry standard approaches to anonymisation of text, such as Azure Language Service [1], use lists of personally identifying information (PII) for removal. Recent research improves the threat model from entity-based detection of PII identifiers to inferable PII, by using an LLM in an adversarial framework. Yet the limitations of PII removal as a basis for anonymisation have been established both empirically [11], and in alignment with privacy legislation

Legislation such as GDPR focuses upon outcome, and not removal of finite lists of identifiers, requiring anonymisation defends against “*all the means reasonably likely to be used ... either by the controller or another person*”. Additionally, personal data is defined as “*any information*” related to an identified or identifiable individual. It is difficult to determine what auxiliary information an adversary may possess, and by using a fixed finite set of identifiers for removal means that any information outside of it, is never anonymised [4].

Consider the following example:

I began my book balancing practise, by placing it on my head. This time I tried a heavier volume with a green and blue spine as my friends watched in disbelief.

This text contains no PII identifiers. However, it may be reasonably likely for one of the friends present, with background information of this strange practise, to read this

text and confidently reidentify the subject. Industry standard methods would therefore fail to prevent re-identification since no standard PII identifiers occur in surface form to remove. Neither is any standard PII identifier inferable from this text, and current state-of-the-art approaches would not prevent re-identification either. This is not just hypothetical. In Figure 1.1 we show a real example where a state-of-the-art method fails, removing names, birthplace and date of birth, but retaining a verbatim description of a distinctive watercolour technique the subject invented.

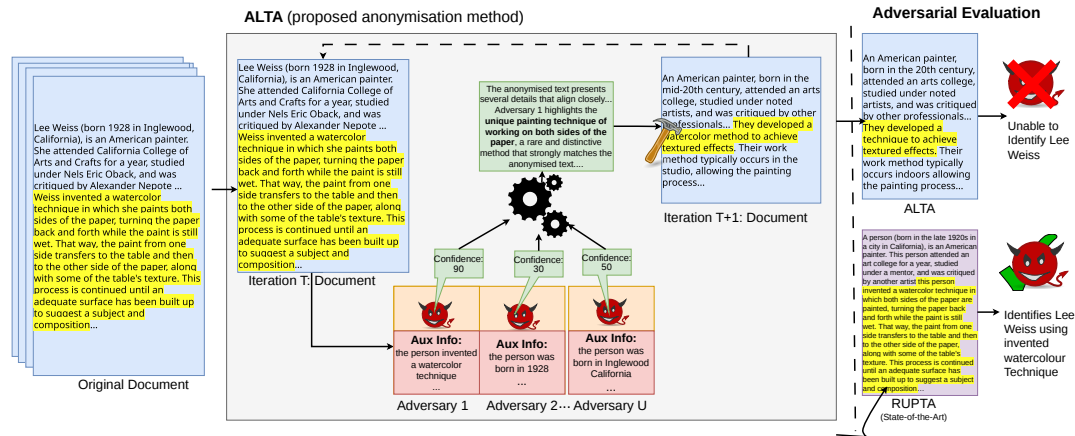


Figure 1.1: We show our main proposed textual anonymisation method, ALTA and compare its anonymised output to state-of-the-art approach RUPTA. The anonymised document is rewritten iteratively by using feedback from an ensemble of adversaries. The invention of a water colour technique matches Adversary 1’s auxiliary information and since this is a highly identifying detail, it is confident it belongs to the same person. This piece of information is then emphasised in a feedback directive used to rewrite the document. Unlike RUPTA, which leaves the description of the invented technique in full, ALTA focuses on generalising the most distinctive pieces of information. This leads to an anonymised document where the adversary can no longer correctly re-identify the individual.

This is not a limitation unique to distinctive hobbies or inventions: any sufficiently large subset of information can allow the reidentification of an individual, even if this information appears innocuous and is not a PII identifier [12]. No individual film rating is personally identifying, yet using only eight film ratings and their approximate dates, Narayanan and Shmatikov [11] show that this uniquely identifies 99% of users in the Netflix dataset of over 480,000 users.

Motivated by the limitations of current state-of-the-art LLM Adversarial frameworks,

we propose changing their adversarial method from inference of PII to linkage of auxiliary information. We use an ensemble of adversaries with disjoint background information from the original text. These adversaries try to link this information the current anonymised document and this feedback is used to rewrite it.

We propose two new anonymisation methods based upon this, **ALTA** (Auxiliary Linking Textual Anonymisation) and **ALTA-FD** (Auxiliary Linking Textual Anonymisation with Fact Distortion), that we find to drastically improve privacy over current state-of-the-art methods, reducing semantic leakage by over 85.3%, whilst incurring only a reduction of 1.3% for task utility. This reduction is despite not having access to the downstream task-label, unlike the method, RUPTA, we compare against.

Existing LLM adversarial methods generalise a document each iteration by rewriting it in a single step, using only the adversary’s feedback and the previous rewritten document. ALTA-FD introduces a fine-grained rewriting process whereby a document is decomposed into a set of facts that are then transformed and rewritten into an anonymised document. Unlike previous approaches that focus on pure removal of information through generalisation, we use both generalisation and perturbation style operations. Perturbation introduces false synthetic information, and unlike generalisation, allows anonymised documents to falsify an adversary’s background information. We discuss the implications of this in Section 4.4.2.

We stress test the utility cost of ALTA-FD by anonymising medical board examination questions where perturbations could easily change the correct inferable answer. Despite this, we observe improvements on both measures for privacy and task-utility over pure ALTA.

State-of-the-art method RUPTA preserves utility by conditioning the anonymisation on the ground truth task label, using the following as part of the prompt:

”You should now improve the utility score about classifying the text as {label} without decreasing the privacy score.”

making the anonymised text tailored for one specific task. To support better data reusability, as desirable for data in research, we design our method to be task-agnostic, finding increases in measures of general utility with only minor task-specific reductions.

In this dissertation, we walk through state-of-the-art LLM adversarial anonymisation approaches, describe and justify our proposed method, present and analyse results on two datasets and discuss the limitations and conclusions of our work. We aim to answer the following research questions:

- RQ(1) Does our proposed adversarial model improve anonymisation privacy?

- RQ(2) Can fact distortions more readily preserve document style and readability
- RQ(3) To what extent does our task-agnostic approach reduce task-specific utility?

These research questions map onto our main contributions:

- A new adversarial anonymisation framework using linkage of auxiliary information to anonymise text that exceeds state-of-the-art methods on both measures of privacy and general utility
- A novel fine-grained document rewriting process using both generalisation and perturbation, that improves privacy and utility over standard single-step rewriting
- A task-agnostic anonymisation method that exceeds state-of-the-art on general utility measures and falls only slightly short of task-utility against existing task-specific approaches.

Chapter 2

Background

In this chapter, we motivate the need for effective anonymisation methods by pointing to the increasing accumulation of personal data and the emerging threat posed by LLMs, which can act as large-scale de-anonymisers. We review current textual anonymisation methods, from the industry standard approach of PII-span removal to newer research that uses LLMs in an adversarial framework. We examine three recent LLM adversarial frameworks, namely Adversarial Feedback, IcoGniText, and RUPTA, and show how they differ in Table 2.1. We then identify three shared limitations of these methods: (1) using PII as the mechanism for anonymisation (2) using only generalisation or perturbation, and (3) RUPTA re-introducing utility for only a single downstream task. These limitations form the basis of our proposed methods ALTA and ALTA-FD that we detail in Chapter 3.

2.1 Requirement For Stronger Anonymisation Methods

Organisations are increasingly accumulating large amounts of personal data whose potential is currently untapped. Under data protection laws such as GDPR, to use this data outside of the reasons it was collected requires a legal basis, such as obtaining explicit consent from individuals, which is often untenable [13]. This restriction is lifted for anonymised data since it is no longer considered personal data. At the same time, the proliferation of large language models enables adversaries to cheaply and effectively de-anonymise unstructured data like text, at a scale not previously possible [16, 7]. High-dimensional data is known to be particularly difficult to anonymise because such data is vulnerable to linkage against small amounts of auxiliary information [11]. Robust anonymisation measures are therefore crucial to both utilise the untapped potential

of this data, whilst protecting against the increasing threat of large-scale automated de-anonymisation.

2.2 Existing Textual Anonymisation Methods

Method	Auxiliary Information	Threat Model	Rewrite Granularity	Operation	Utility Mechanism
Azure Language Service [1]	None	Surface identifiers	Span	Removal	None
Adversarial Feedback [17]	None	Attribute inference	Document	Generalisation	None
IncogniText [3]	None	Attribute inference	Document	Perturbation	None
RUPTA [20]	None	Attribute inference	Document	Generalisation	Task-conditioned
ALTA	Derived facts	Linkage	Document	Generalisation	Task-agnostic
ALTA-FD	Derived facts	Linkage	Fact	Both	Task-agnostic

Table 2.1: Characterisation of existing text anonymisation methods against our proposed methods. *None* under auxiliary information indicates that the method does not model an adversary holding background information, treating identifying attributes as sufficient for re-identification on their own. *Derived facts* are facts extracted from the source document. ALTA and ALTA-FD are described in full in Chapter 3.

We categorise relevant existing methods for anonymisation of text in Table 2.1 to allow us to discuss and compare them. We include industry standard Azure Language Service, since it serves as a span-based baseline that LLM-adversarial methods benchmark against [16, 17, 3, 20]. We include the three LLM-adversarial frameworks for anonymisation that we discuss throughout the rest of this thesis, which, to the best of our knowledge, were the first three methods proposed using this framework. Namely Adversarial Feedback, IncogniText and RUPTA.

Span-based methods for the anonymisation of text, such as Azure Language Service [1] or Microsoft Presidio [10], have been shown to be ineffective at protecting privacy, both in terms of recall (masking all PII spans), precision (masked spans being correct), and in inferences of PII from non-explicit cues in the text (we give an example to illustrate what is meant by inferrable PII in Section A.3) [8, 16].

Adversarial Feedback is the framework that first proposed adopting LLMs into an adversarial framework to anonymise text, following previous work that found LLMs were comparable with humans in making inferences of personal identifiers [17, 16]. In this framework, one LLM acts as an adversary and the other as an anonymiser. The framework iteratively anonymises a document, with the adversary’s inferences and guesses of PII attributes being supplied to the anonymiser to rewrite the text. This rewritten text is given back to the adversary to make PII inferences again, with this

cycle continuing for a fixed number of iterations. Adversarial Feedback has been shown to exceed commercial methods substantially on measures of privacy under an attribute inference threat model.

IncogniText builds upon this iterative adversarial framework, but instead of generalising information related to identifying attributes, it introduces false synthetic information to mislead an adversary into guessing false randomly selected personal attributes [3]. Another framework, RUPTA, builds upon Adversarial Feedback, focusing on retaining the utility of the anonymised text for a downstream task. When the anonymised document exceeds a privacy threshold, anonymisation switches from a privacy rewriting mode to a utility rewriting mode that tries to improve classification of the text using its ground truth label [20].

As noted in the limitations of Adversarial Feedback, their framework does not guarantee that the text is anonymised against an adversary with additional background knowledge [17]. Since all three of these approaches centre on removal of PII, they do not assume an adversary to have any background information and that from the inference of PII alone, re-identification can occur. Our proposed methods, ALTA and ALTA-FD, assume an ensemble of adversaries, each with differing background information that they use to link to the anonymised document. Unlike previous methods, ALTA-FD introduces rewriting at the fact-level, either through generalisation or perturbation (we give examples of what we mean by these operations in Section 2.4.2). Like RUPTA, we inherit a utility reintroduction mechanism, but unlike them, we do not condition it for improvement for a single task, but try to improve utility more generally.

2.3 Textual Privacy Evaluation

There is no agreed upon consensus for evaluating the privacy of anonymised text. Xin et al. [19] introduce a privacy evaluation framework using a reconstruction style attack, where an adversary attempts to link background information, to an anonymised document. They then measure the semantic distance between the original document and the linked anonymised one, with greater distance indicating greater privacy. Under this framework, they find current state-of-the-art methods, including Adversarial Feedback, to produce texts substantially less private than under an attribute inference definition. The authors call for anonymisation methods that protect against semantic information leakage and to our knowledge, no LLM-based anonymisation methods use a linkage adversary in its rewriting loop to achieve this, which is what we propose in this thesis.

2.4 Limitations of Existing Methods

We discuss the three main limitations we find of existing methods, (1) using PII as the mechanism for anonymisation (2) using only generalisation or perturbation, and (3) re-introducing utility for only a single downstream task. Our proposed methods cleanly attend to each of these limitations; ALTA uses a linkage based adversary with a task-agnostic utility re-introduction mechanism. ALTA-FD does this but also allowing both generalisation and perturbation.

2.4.1 Insufficiency of PII anonymisation

Consider the following example:

|| *They were the first person to run a four minute mile.*

This text contains no standard PII identifiers, and yet it precisely identifies a single individual. Although Bannister is a public figure and re-identification presents no harm here, it could for non-public individuals where distinctive, non-PII background information is held by others. Recall the book-balancing example from Chapter 1 which contains no standard PII identifiers but if shown to the friends present would almost certainly reidentify this individual.

The limitations of PII identifier removal for anonymisation are not only confined to toy examples, but have been demonstrated on public large-scale datasets, with real-world implications. Narayanan and Shmatikov [11] in their seminal paper, deanonymised the Netflix dataset¹ using only publicly available data from IMDB. This was despite the released dataset having PII removed. They later argue that **any** attribute can be identifying in combination with others, giving the example of books someone has read or clothing in their wardrobe; individually, each item does not identify the individual, but for a sufficient number of items, it has the potential to do so [12]. Text is particularly difficult to anonymise, due to its inherent high dimensionality. Unlike structured data that contain a finite number of attributes, the number of propositions that can be made about a subject is unbounded and so is the amount of specificity.

Not only is there an empirical precedent to improve anonymisation from PII removal, but a legal one. GDPR Recital 26 asks whether a person can be identified using “*all the means reasonably likely to be used*” [2]. This is a question of outcome, what could

¹The eponymous ‘Netflix Dataset’, was a dataset released by Netflix containing film reviews, released as a challenge to build recommender systems.

an adversary achieve, not about which categories of data are present. This includes our second example, where regulatory guidance confirms this is not limited to publicly available information but includes acquaintances who “*know a great a deal about another person*” and may be able to identify them “*even if another member of the public would be unable to*” [5]. A document can have every enumerated identifier removed and yet still fail this standard as in the case of our two examples.

Despite these known limitations, removal of PII remains the main industrial method of anonymising data. Recent research has moved past entity-recognition-based detection of PII, to PII inferrable by an LLM (see A.3 for what is meant by inferrable PII). [17, 3, 20]. Although this offers an improvement from surface level detection, such methods fail to anonymise real examples containing no direct or inferable PII (see Figure 1.1).

2.4.2 Methods use Generalisation or Perturbation but Not Both

Generalisation is where a piece of information is replaced with a strictly more general version (e.g. “a lightning-bolt scar on his forehead” → “a scar on his forehead”). Perturbation is where some or all of a piece of information’s entities or values are replaced, such that the original is no longer true (e.g. “he liked cats” → “he liked dogs”). Generalisation allows statements to remain true but less specific, whereas perturbation retains specificity at the cost of truth.

Current LLM adversarial-based methods commit to a single rewriting operation type. Adversarial Feedback and RUPTA generalise cues to identified PII, whereas IncogniText rewrites the document to deceive an adversary into guessing false PII. To the best of our knowledge, no existing LLM adversarial method explores using both. We compare each operation on a single example to illustrate the tension between specificity, truth and the inferences that can be made:

- (1) **Original.** *Molly was 23 years old when she received the 12th point on her driving licence.*
- (2) **Generalisation.** *A young adult accumulated multiple penalty points on their driving licence.*
- (3) **Perturbation (sound).** *Samantha was 26 years old when she received the 15th point on her driving licence.*

- (4) **Perturbation (unsound).** *Samantha was 26 years old when she received the 10th point on her driving licence.*

Surface Form. Compare (2) with (3) and (4). Generalisation, (2), makes the text vaguer and structurally unlike the original. Molly, 23, becomes “a young adult”, her gender is removed, and a specific count becomes “multiple”. The perturbed versions retain an explicit name, age, and count and read much more alike the original in form. The anonymised text could be used for training a model, to prevent leakage of personal data or for people to learn from. In both cases, the anonymised data is out of distribution of the original, and so we incur a utility cost from the divergence in both form and style. We provide lengthier examples of the improved surface form from using perturbations in Figure 4.1 and Figure 4.2.

Contradictions. Suppose an adversary with background information of a 23 year old on the 12th point on their licence. (2) is consistent with this background information, so the adversary’s confidence is reduced but never eliminated. Version (3) and (4), on the other hand, contradict the adversary’s background knowledge, since the number of points differ from it. Generalisation can only make an adversary with accurate background information less certain, but can never contradict it.

Inferences. In the United Kingdom, accumulating 12 or more penalty points on a driving license results in disqualification. Variant (2) destroys this inference, since “multiple” could both refer to above or below this threshold. Variant (3) retains this inference since 15 is contained within the same equivalence class as 12, of being disqualifying. Variant (4) does worse than destroy the inference, but instead leads to a false one: the driver is not disqualified. A sound perturbation, where this inference is preserved, improves over a generalisation where it is lost, but which improves upon a unsound perturbation, where an incorrect inference is confidently drawn.

From our discussion of these two operations we devise the following rule: perturb a piece of information if we can confidently find a replacement where the same general inferences of the document can be made, otherwise generalise. Applying these rules requires reasoning about each piece of information or fact individually, since we established in Section 2.4.1 that the information requiring treatment is not limited to PII and that any piece of information can lead to re-identification. Anonymisation at the fact level, using both operations, therefore follows from both of these limitations.

We adopt this fact level rewriting approach with both generalisation and perturbation operations in ALTA-FD. This allows us to report on the effects of a mixed fine-grained approach against the more typical full document generalisation adopted by ALTA,

RUPTA and Adversarial Feedback (see Table 2.1). Specifically, we can discuss whether utility improves, through preserved surface form and correct inferences, and privacy improves through contradictions with adversary’s background information.

2.4.3 Reintroducing Utility via a Single Downstream Task

RUPTA has both a privacy and utility mode. In its privacy mode, it focuses on anonymising the text against a specific PII attribute. Upon exceeding a privacy threshold, it switches to a utility reintroduction mode where it improves utility by prompting an LLM to improve classification of a ground-truth label for a single task. We note three specific issues with RUPTA’s single-task implementation.

First, is data reusability. In many cases, anonymisation is done at data release time, before knowing all downstream uses of it. For research, released data is most useful where it can support reuse for many downstream tasks *beyond* those the publishers conceived. It may therefore be beneficial to anonymise without assuming a known single downstream task at anonymisation time for data reusability.

Second, it requires your original data to be labelled with the ground-truth labels for the single task you want to apply it to. Even if the task is known, these labels may not be available at anonymisation time. For example, it may be that you can only legally obtain the labels *from* the anonymised data, making it impossible to use the labels as part of the anonymisation, where they do not yet exist.

Third, using the ground truth-label during anonymisation is a form of leakage if utility is evaluated with a downstream task using those same ground-truth labels. For example, on DB-BIO, where the downstream task is to predict occupation labels, the anonymiser is supplied the text’s ground-truth label whilst anonymising it. This means the ground-truth labels are used as both part of the anonymisation process, to reintroduce utility, and as a ground truth for evaluation. We question whether providing such information during anonymisation ensures that the downstream task still offers a reliable measure of utility. For example, this leakage could lead to the label being reintroduced verbatim to the text while all other useful information is removed.

From the limitations of RUPTA, we believe there is a need for anonymisation that is task-agnostic. This would allow the benefits of reusable data without needing ground-truth labels and the risk of using them for anonymisation.

Chapter 3

Method

The three main objectives of our anonymisation system are to anonymise against an adversary with information outside PII, allow anonymisation through both generalisation and perturbation, and reintroduce utility in a task-agnostic fashion. These objectives address each of the three limitations we identify in our background in Section 2.4. In this chapter, we lay out the principles behind our approach, present our methods, **ALTA** and **ALTA-FD**, and describe the datasets and metrics we use to evaluate them.

3.1 Overview

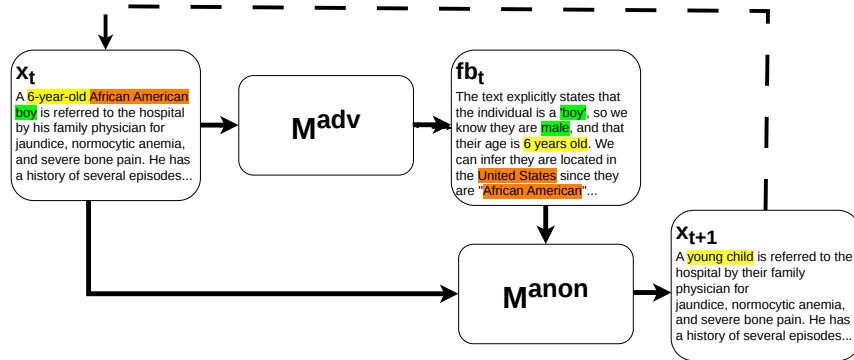


Figure 3.1: The core pipeline used by Adversarial Feedback, IncogniText and RUPTA for LLM-based anonymisation. The current iterations text, x_t , is provided to an LLM that acts as an adversary, M^{adv} , by inferring PII in the text and providing feedback fb_t . This feedback is used by M^{anon} to rewrite x_t into x_{t+1} and better preserve the individual's privacy.

Our anonymisation method is in the same family as other adversarial LLM-based

methods such as Adversarial Feedback [17], Incognitext [3] and RUPTA [20]. Like them, our main pipeline is to iteratively rewrite a document using the feedback of another LLM that acts as an adversary. We show the core pipeline that all such methods use in Figure 3.1.

We build upon RUPTA’s codebase, making this the closest method to ours. RUPTA extends the core pipeline in Figure 3.1 with a utility mode for improving performance on a downstream task. During anonymisation, each document is scored on both privacy and utility. Once the privacy score exceeds a threshold, the document is considered to be no longer identifying, and it is instead rewritten to improve utility performance on a downstream task. In their paper, the downstream task is a classification task and the ground-truth labels are used to guide these utility improving rewrites.

From RUPTA, we inherit the privacy utility threshold mechanism. Our method differs from RUPTA in three major ways, to address each of the three limitations we identified:

1. **Adversarial Model.** For all prior methods, the adversary, shown as M^{adv} in Figure 3.1, attempts to infer PII from the current document and the anonymiser model uses the adversary’s feedback on what it could infer to better anonymise it. We introduce an adversary that tries to link randomly selected atomic facts from the original document to the current anonymised one. These randomly selected facts act as auxiliary information granted to the adversary and remain fixed throughout the rewriting process, standing in for the background knowledge someone could reasonably possess. The adversary provides feedback on how confident it is that an individual with these facts is the individual described in the anonymised text. This confidence feedback is synthesised into a single directive, which is then used to better anonymise the text.
2. **Document Rewriting.** The process by which we rewrite the document to include both generalisation and perturbation. Current approaches use the feedback from the adversary to rewrite the previous document in one go. We introduce a fine-grained rewriting process whereby the document is decomposed into a set of atomic facts that each undergo a transformation. We define transformations that generalise or remove information (coarsen, delete), that invent new information (perturb, specialise) and that preserve it (identity). Each transformation is then applied with a strength of low, medium or high. A description of each transformation type and strength is in Section A.4.3.2.1. The next anonymised document

is reconstructed using these transformed facts. Specifically, we make a separate LLM call to write a naturally flowing document using the list of transformed facts. This entire rewriting process, we coin *fact distortion*.

3. **Utility Reintroduction.** The process by which we re-introduce utility. RUPTA uses the ground truth label for a specific downstream task to reintroduce utility. We do not assume such a task is defined at anonymisation time and instead devise a general method of reintroducing utility. Specifically, our method improves utility of the current anonymised document by making conclusions and inferences made from the original more clear.

We propose **ALTA** (Auxiliary Linking Textual Anonymisation) and **ALTA-FD** (Auxiliary Linking Textual Anonymisation with Fact Distortion).

ALTA specifically incorporates our changes to the **(1) Adversarial Model** to use document derived auxiliary information and **(3) Utility Reintroduction** which uses the inferences and conclusions from the original to reintroduce utility to the anonymised document. Document rewriting is done in RUPTA fashion by rewriting it in one step and only by generalising information using the adversary feedback.

ALTA-FD, like **ALTA** incorporates changes **(1)**, **(3)**. It also adds our change **(2) Document Rewriting** to rewrite documents using the adversaries feedback in a multi-stage process, where facts from the current document are transformed by both generalisation and perturbation style operations.

3.2 Design Rationale

Under the umbrella of these three main changes we make several other decisions which we wish to state and justify clearly.

3.2.1 Adversarial Model

Adversaries that match auxiliary information. Each adversary attempts to match auxiliary information to the anonymised document. This is distinct from existing adversarial frameworks which instead give feedback on inferrable PII (see Table 2.1). We justify why this approach is more legally aligned in Section 2.4.1.

For a piece of text, we ask the following question about each piece of information: if an adversary *were* to know this piece of information about an individual, how confident would they be, upon reading the document that this is the **same** individual.

Consider the following:

I live near the stadium whose song is 'You'll Never Walk Alone', and invented a watercolour technique in which both sides of the paper are painted, turning the paper back and forth while the paint is still wet.

Adversaries that infer PII, if location was part of their identifiers, would be able to infer that the person lives in Liverpool, United Kingdom, and their feedback would anonymise the document to remove this inference:

I live near a stadium with a distinctive song and invented a watercolour technique in which both sides of the paper are painted, turning the paper back and forth while the paint is still wet.

However, unless some version of “invented a technique” or “notable achievements” is provided as a PII attribute, details of inventing the water colour technique would not lead to the inference of PII, and thus remain preserved. Figure 1.1 shows this occurring in practice, where RUPTA’s output retains the technique in identifying detail.

For our method, the information of living near the stadium, and of inventing a water colour technique would form auxiliary information known by some adversary. Our adversarial feedback would therefore flag both the location cue and the watercolour technique as highly identifying, and both would be anonymised.

This allows us to capture identifying information outside of that which can be caught by fixed PII based methods, that a priori must provide a finite list of identifiers.

Adversaries with Document Derived Knowledge. Every anonymisation method must make an implicit choice about what auxiliary information it assumes an adversary to have. Existing methods assume an attacker with no auxiliary information and that from extraction of PII alone, the adversary would reidentify the individual. We assume a stronger attacker, who has auxiliary information derived from the original document. More specifically, we give a subset of atomic facts from the original document to the adversary to act as its auxiliary information. As we demonstrated in Chapter 1, this means we can detect and anonymise against identifying information outside of PII, which current methods fail to.

It is worth noting that real adversaries can have auxiliary information outside of that which is document derivable. However, this is a limitation shared by all anonymisation

methods, unless additional auxiliary information is supplied during the anonymisation process. We discuss this further with an example in our limitations, in Section 4.4.2.

Using an ensemble of adversaries. We divide all possible auxiliary information between U adversaries. This allows us to use all pieces of information from the original document, whilst simulating adversaries with differing background knowledge. We believe this is more faithful to real attackers that could range from strangers, friends or colleagues. In a similar way, using an ensemble of attackers allows us to view if the document is sufficiently anonymised from differing perspectives. We summarise the feedback across all adversaries, highlighting the most identifying information that is in need of being changed for anonymisation.

Privacy as a worst-case scenario. We view privacy as a worst-case re-identification scenario; if just one adversary has the requisite background information to re-identify the individual in the document, it is not anonymised. This is consistent with the ICO’s guidance to assume a “determined person with a particular reason to want to identify individuals. For example, investigative journalists, estranged partners, stalkers, or industrial spies.” Each of these individuals may have differing background information.

3.2.2 Document Rewriting

Any identifying information should be rewritten. Following from the design of our adversary, which focuses on all information in the document allowing re-identification, we allow **any** piece of information in the document to be rewritten. This is broader than previous methods that only rewrite information in the document related to the inference of PII.

Generalisation and Perturbation are used. Generalisation style anonymisation alone can make documents vague, and unlike the original in form. Perturbation style operations retain surface form better but can lead to false inferences being made (we discuss this in depth in Section 2.4). We believe such types of operations are complementary.

Anonymisation should be at a fact level. We want to be able to control at a fine level the information in the text which is altered, how it is altered (generalisation or perturbation), and by what strength. Current methods use the adversarial feedback to rewrite the document in one go. To improve the level of control an LLM has and to force it to consider each piece of information, we deconstruct a document into a list of facts from which anonymisation is applied. This allows the LLM to apply the most appropriate type of transformation for each piece of information individually. We

believe this allows greater manipulation over the document.

3.2.3 Utility Preservation

Anonymisation is lexicographic optimisation. The privacy-utility trade off is a well-known property of anonymisation where increased privacy leads to a reduction in the utility and vice versa. We inherit from RUPTA their framing of anonymisation as a lexicographic optimisation task, where privacy is given the higher priority, followed by utility. In other words, two anonymisations are judged based upon their privacy, and only then their utility if their privacy level is equal. Following from this, we optimise the privacy of our texts until we exceed a privacy threshold τ , at which we point we consider our texts effectively anonymised. We then switch to a utility re-introduction mode, moving back to privacy mode if the privacy score ever falls below τ .

Anonymisation should carry the same inferences and conclusions. We believe optimising a version of utility defined by if the same inferences and conclusions can be drawn from the anonymised document is sensible. Even if individual details are lost or changed, would the reader have the same general idea? If the same inferences can be drawn, then different types of questions or tasks about the anonymised text should still yield the same results as the original. We believe optimising for this task-agnostic definition of utility is more in line with our desire to create reusable data, as opposed to a RUPTA single task approach.

3.3 Proposed Method

In this section we describe our anonymisation methods ALTA and ALTA-FD through a worked example. We provide a more precise, but less accessible formal description of both methods in our Appendix at Section A.4.

3.3.1 ALTA

We explain ALTA through a worked example for the first iteration in Figure 3.2. The original document is decomposed into a set of atomic facts shown by F_{orig} , which are divided equally between the adversarial models, acting as their auxiliary information throughout anonymisation.

ALTA (one iteration, $t = 0$)

Example demonstrating anonymisation for a fictional person

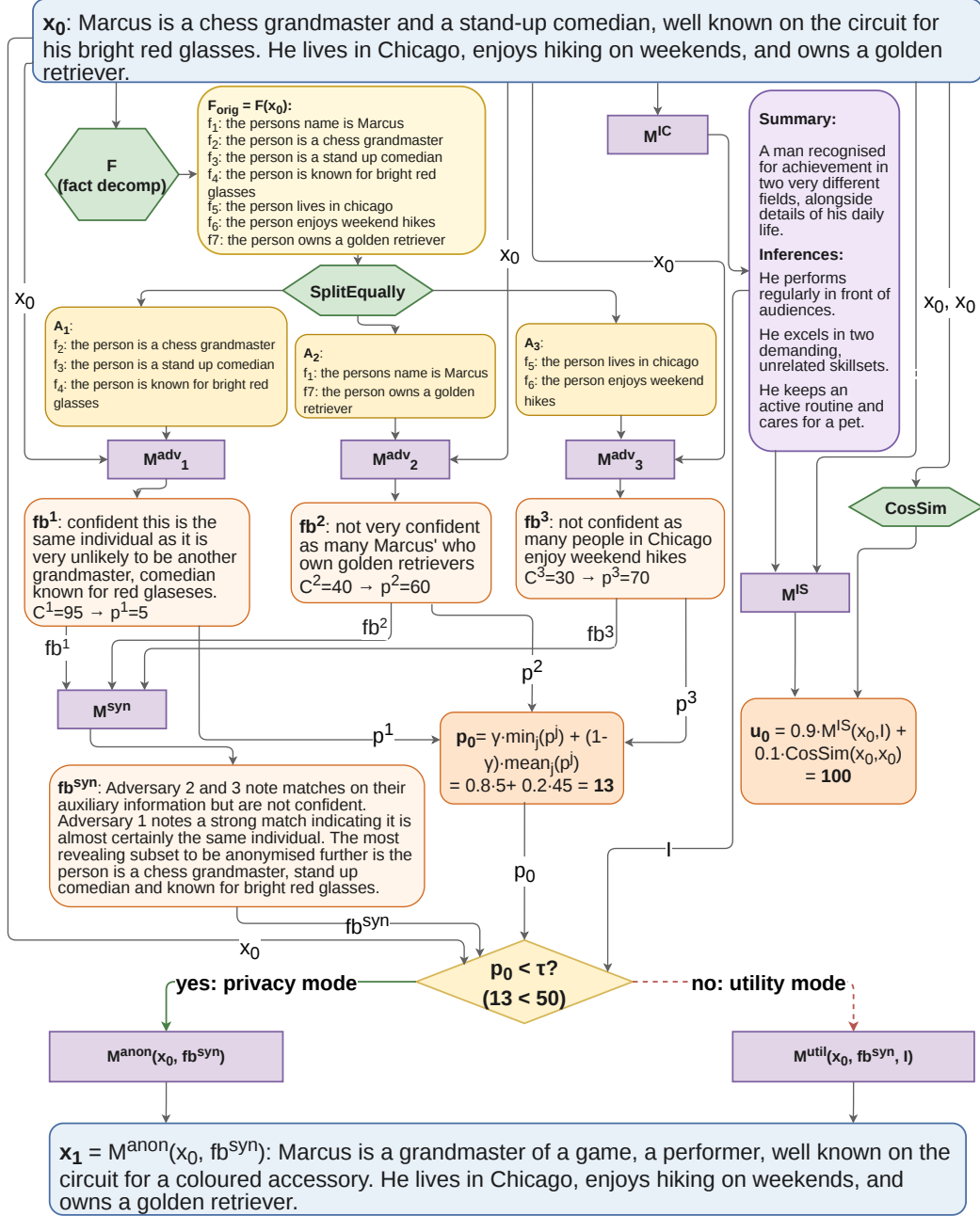


Figure 3.2

For privacy evaluation, each adversary j provides feedback fb^j and a confidence score C^j indicating how confidently they believe the current anonymised document matches the background information. Since this is the first iteration, the current anonymised document being evaluated *is* the original document and so all the details match. Even still, wherever details match with an adversary’s auxiliary information, it considers how identifying the details it matches on are. In our example, matching on being a grandmaster, comedian and with red glasses is far more identifying than matching on the person living in Chicago and enjoying weekend hikes, which could describe many people.

Feedback from the j adversaries is synthesised into a feedback directive fb^{syn} , and a single privacy score is computed from each adversary’s privacy score P^j . Following from our Privacy as a Worst case design in Section 3.2.1, we let the combined privacy be a weighted sum of the minimum and mean privacy across our adversaries. We use the weighting 8:2.

The utility of the current anonymised document is computed using a weighted sum of an LLM judged score of how well the current document follows the originals conclusions and inferences and its cosine similarity with the original document. For iteration 1, since the current anonymised document *is* the original document, the cosine similarity will be 1, and the inferences and conclusions will perfectly follow.

After obtaining a privacy and utility score for the document, it is rewritten. If the privacy is below the privacy threshold τ , we rewrite in privacy mode, using the current document and the synthesised feedback from the adversaries. If the privacy is above τ , we consider it effectively anonymised, and reintroduce utility, by rewriting it to better reflect the inferences and conclusions made from the original document I .

3.3.2 ALTA-FD

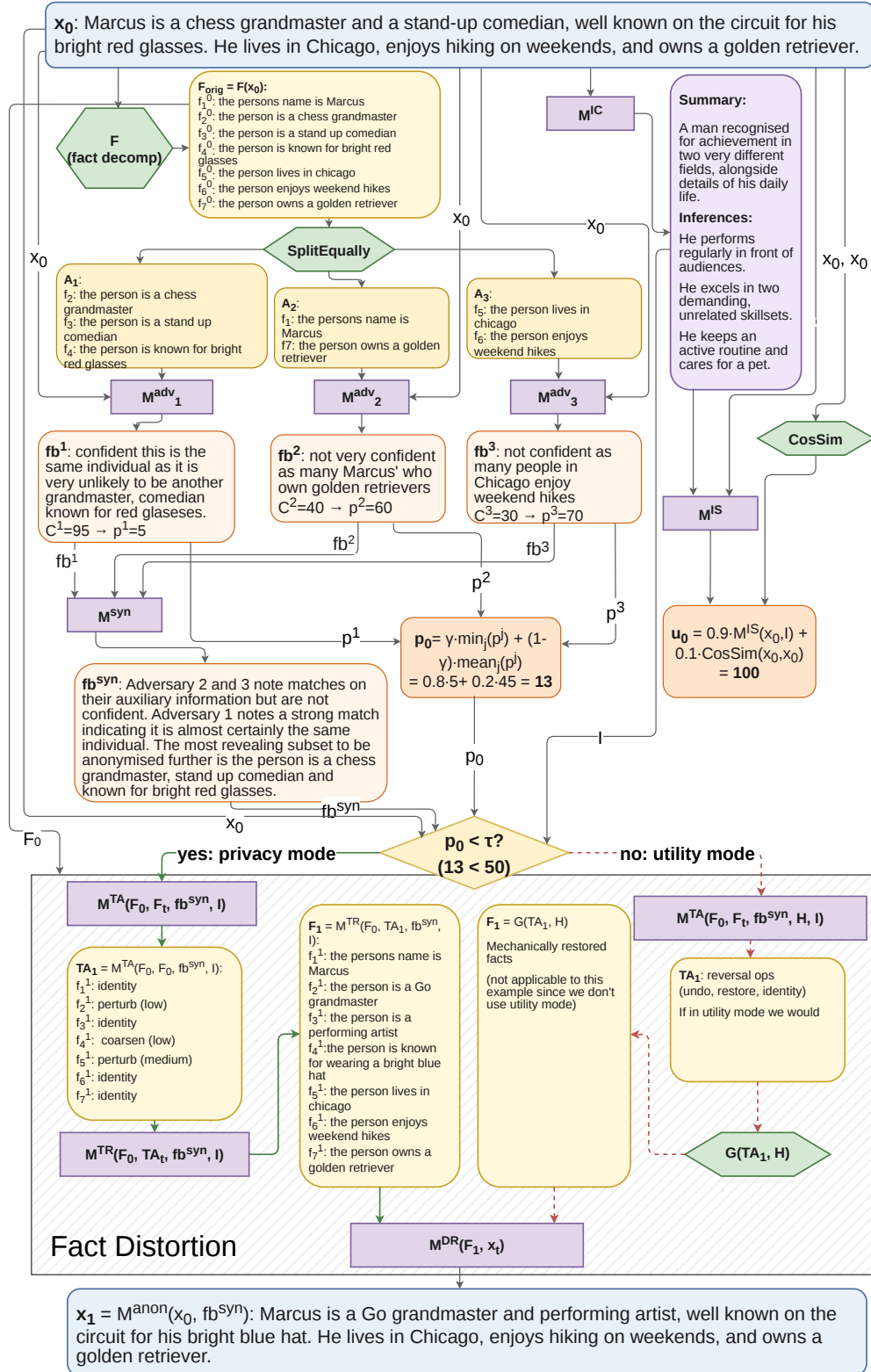
We explain ALTA-FD also through a worked example for the first iteration in Figure 3.3. The only difference from ALTA is the process by which the document is rewritten. Everything up until then is the same as before.

The current anonymised document is below privacy threshold $\tau = 50$ and we therefore enter privacy mode as before. Since we are at iteration $t = 0$ the current document is the same as the original document x_0 .

This time an LLM M^{TR} selects a fact transformation type and strength to apply to each fact in the current document. This can be selected from generalisation style

ALTA-FD (one iteration, $t = 0$)

Example demonstrating anonymisation for a fictional person



x_1, F_1 becomes input to iteration $t=1$

Figure 3.3

operations (coarsen, delete), those that insert new information (perturb, specialise), or none (identity). The strengths range from low, medium to high.

To apply appropriate transformation types and strengths we pass in the synthesised adversarial feedback fb^{syn} and in our prompt provide guidance to the LLM for transformations assignment. Optionally, the inferences and conclusions from the original document can be passed in to more strongly emphasise these should be preserved when selecting the transformations.

These assigned transformations, TA_1 are then applied by passing them, the current facts F_0 and the synthesised adversarial feedback fb^{syn} to LLM M^{DR} . This LLM is responsible for transformation rewriting, and follows guidance on rewriting each fact according to its assigned transformation. Again, we optionally allow the inferences and conclusions from the original document to be passed in to ensure the rewritten facts preserve them.

These rewritten facts are then reconstructed into the next iterations document, x_1 . For the next iteration, the current document becomes x_1 and the current facts becomes the distorted facts F_1 .

3.4 Datasets

We choose two well known datasets, DB-BIO and MEDQA, both of which have been used to benchmark privacy and utility for previous textual anonymisation methods.

3.4.1 Overview

Dataset	Domain	Text Type	Size	Avg. Length (chars)	Min Length	Max Length
DB-BIO (test)	Celebrity	Biographies	239	1402	825	3226
MEDQA (subset)	Medicine	Patient Summaries	250	660	82	2396

Table 3.1: Statistics from the two datasets we use. DB-BIO, consists of celebrity biographies and on average is over twice as long as MEDQA, which consists of medical profiles of patients outlining their symptoms and background.

3.4.2 DB-BIO

Description DB-BIO consists of medium length celebrity biographies taken from Wikipedia. We run our anonymisation on the test set, which consists of 239 biographies from celebrities spanning 24 occupations. DB-BIO was introduced by Yang et al. [20] in the same paper in which they introduce RUPTA, acting as their primary dataset for evaluation.

Privacy. Yang et al. [20] produce a dataset containing celebrities, as they reason an LLM would be able to act as an adversary with background information by using its pre-trained knowledge to guess the celebrity from the anonymised text. The main privacy evaluation metric they use is *Success Rate*, which is the percentage of correct guesses an LLM makes when selecting the celebrity from a list of similar distractors.

Task Utility. We assess task utility as being able to accurately guess a celebrity’s occupation from their anonymised biography. We use the occupation classifier Yang et al. [20] provided for this task, that is trained using non-anonymised biographies. This is an easy task, as reflected by the near-perfect results in both our results in Table 4.1 and the author’s, since the biographies typically contain the celebrities occupation both in its surface form and is referenced throughout.

Dataset Suitability. We believe DB-BIO is a suitable dataset to demonstrate our anonymisation methods for several reasons. First, it allows us to draw a comparison against our model’s closest ancestor, RUPTA, since this is the primary dataset they use. Second, celebrity biographies contain authentic personal information about natural individuals. We considered alternative synthetic datasets, but these are generated using simplistic personas and would lack the diversity or realism of data from real people. Thirdly, it provides a natural alternative privacy metric that we can compare to semantic distance, our main privacy metric. For this evaluation, a stronger LLM is used than for anonymisation to simulate an adversary with greater auxiliary information and reasoning.

3.4.3 MEDQA

Description. MedQA consists of multiple-choice questions from the US medical licensing exam to assess medical understanding and reasoning skills. There are 11,450 questions that consist of fictitious patient profiles, followed by multiple-choice questions with 4-5 possible answers. We randomly sample 250 questions from the original dataset and filter out questions not describing an individual.

Privacy. Patient profiles contain both standard PII and rare identifying combinations of symptoms and background information. We believe this provides a realistic test of privacy, since a patient can be reidentified through both PII and innocuous information whose conjunction is rare. We use semantic leakage as our privacy metric, which was applied to MEDQA in the paper which proposed the evaluation metric [19].

Task Utility. MEDQA has a clear downstream task of correctly answering the question about anonymised versions of the patient profiles. We measure task utility by adopting how previous work does so, where an LLM takes the test using the anonymised profiles, and we validate its answers with the answer labels [19, 17, 9]. The degradation in performance on the test accuracy from taking the test using the original, unanonymised profiles allows us to quantitatively measure the effect of anonymisation on utility.

Dataset Suitability. We believe MEDQA is a suitable dataset for evaluating anonymisation for several reasons:

Firstly, there is a clear tension between removing personal information disclosed about the patient in the text, whilst still being able to infer the correct answer. It has also been used previously as a benchmark for anonymisation in several other works [19, 17, 9].

For ALTA-FD specifically, we believe it provides a strong test if fact distortion anonymises texts in a way faithful to the original. Unlike Adversarial Feedback, RUPTA, or ALTA, which anonymise by making information more general, ALTA-FD perturbs information in the text by replacing it with synthetic information. Uncareful perturbations of symptoms, patient background or history could easily change the medical diagnosis that is inferable from the profile, and thus reduce utility. However, careful perturbations could allow the correct inferences to be stronger than generalisation and still protect the individuals privacy. For a more concrete example, please refer to our example on points on a driving license in Section 2.4.2.

As stated in our design decisions in section 3.2.3, both ALTA and ALTA-FD are task agnostic. We therefore do not provide knowledge of the task, the question, the multiple-choice options, the correct answer, or specific prompts written for anonymisation in the medical domain. Both methods are **only** provided with the patient profile and asked to anonymise them.

We believe that MEDQA provides a difficult challenge for ALTA-FD, due to the difficulty of perturbing information without knowledge of which information will be later useful for answering the question. Unlike RUPTA, which, in utility mode, can access the ground truth label and rewrite using this, we only provide ALTA and ALTA-

FD with inferences and conclusions obtainable from the original document to ensure the anonymised document preserves these.

3.5 Experimental Design

We run experiments on our datasets using the following anonymisation methods in the table.

Method	Datasets	Anonymisation
AF [17]	MEDQA	PII inference, generalisation
RUPTA [20]	DB-BIO	PII inference, generalisation, task-specific utility
ALTA (ours)	Both	Auxiliary linking, generalisation
ALTA-FD (ours)	Both	Auxiliary linking, fact distortion

Table 3.2: The anonymisation methods we evaluate.

We test RUPTA exclusively on DB-BIO due to its requirements for the dataset to be accompanied by labelled PII with ground-truth values, which is unavailable for MEDQA. Additionally, RUPTA’s privacy evaluator is designed to anonymise clues against a single PII identifier. DB-BIO works because the identifier is the celebrity’s name, which an LLM can recognise from background knowledge it already holds. For non-famous individuals, no such knowledge exists, so RUPTA can only anonymise clues to the single labelled attribute. Both requirements limit its general use.

For MEDQA, we instead apply Adversarial Feedback (AF) to compare against. This has been applied to MEDQA previously by both the authors and in other work [17, 19]. Consistent with Xin et al. [19]’s evaluation, we apply AF with its full set of personal identifiers to anonymise income, age, gender, location, place of birth, education, occupation, and marital status. Some of these are less applicable to patient profiles; however, Staab et al. [17] detected the presence of each during their evaluations. We test Adversarial Feedback exclusively on MEDQA, since RUPTA is already shown to exceed it on DB-BIO for measures of privacy and utility.

3.5.1 Privacy

No consistent standard exists for evaluating the privacy of anonymised text. Ren et al. [15] survey current approaches to privacy evaluation of anonymised text, grouping

metrics into six categories. They identify reconstruction attacks, where an adversary tries to reconstruct the original text from the anonymised version, as among the most aligned with GDPR. We use Xin et al. [19]’s framework as our primary privacy metric because it is a form of reconstruction attack, and because it uses the threat model that we argue in Section 2.4.1 is better aligned with data protection legislation, and which captures re-identification risk that is not possible under a PII-inference model. DB-BIO offers an alternative strong measure of privacy by *success rate* of an LLM in identifying celebrities from their anonymised text. This was the primary privacy measure used in RUPTA, our method’s closest ancestor, and so we retain this and the LLM’s confidence when prompted with the correct celebrity, *confidence score*. MEDQA, unfortunately, does not come with a simple way of measuring privacy, since it is not annotated for PII. We therefore only apply our main privacy metric, which has previously been applied to it [19].

3.5.1.1 Success Rate ↓ (DB-BIO only)

Measures the proportion of correct guesses when identifying the ground-truth celebrity from a list of 10 candidates (one correct, nine similar distractors). Importantly, the nine distractors are generated by an LLM and are **not drawn from the dataset**; they are plausible celebrities that the anonymised document *could* refer to. The candidate list and the anonymised document are then presented to another LLM, which must select the correct celebrity. This is repeated independently for each anonymised document, and the success rate is the proportion of correct guesses. The metric tests whether the anonymised biography remains identifiable among similar individuals. We include this measure because it was previously used by RUPTA on this dataset.

3.5.1.2 Confidence Score ↓ (DB-BIO only)

LLM-judged confidence (0–100) in identifying the ground-truth celebrity from the anonymised biography. This captures how certain an adversary is, not just whether they succeed, and we include this measure because a dataset can have a low success rate but still leave the adversary highly confident. Confidence is measured by an LLM that is given the anonymised text and the ground-truth celebrity label and asked to rate its confidence that the anonymised text describes the celebrity. Importantly, it is not given the candidate list used for success rate, and only asked to consider its confidence for the true celebrity. We include this measure because it was previously used by RUPTA on

this dataset.

3.5.1.3 Semantic Leakage (DB-BIO and MEDQA)

Xin et al. [19] introduces a privacy evaluation framework using a reconstruction-style attack, measuring re-identification risk for released anonymised datasets.

They consider an adversary with:

1. A released anonymised dataset $\mathcal{D}_{\text{anonymised}}$ that could for example be released for research purposes or leaked
2. Auxiliary information $\tilde{x}^{(i)}$ about each individual i , that for example could be obtained through social media or by a personal acquaintance, given by three randomly selected facts from the original document.

The adversary's goal is linking each individual's auxiliary information to their anonymised record across the entire anonymised dataset to infer sensitive information. The attack is broken into two stages:

Linkage. The adversary uses an individual i 's auxiliary information, $\tilde{x}^{(i)}$, and the anonymised dataset, $\mathcal{D}_{\text{anonymised}}$, to link it to the record they suspect belongs to i , given by $\hat{y}^{(i)} \in \mathcal{D}_{\text{anonymised}}$. This can be represented by a linkage function $L(\tilde{x}^{(i)}, \mathcal{D}_{\text{anonymised}}) \rightarrow \hat{y}^{(i)}$. This linked record is not guaranteed to be the **correct** anonymised record for individual i , but the one selected by the adversary as most likely theirs.

Leakage. After linking to the document they believe is their target's, the adversary tries to learn as much new information about them as possible. The new information that can be learned or *leakage* is measured using the semantic distance between the linked document the adversary believes is the target's, $\hat{y}^{(i)}$, and the original document of the target $x^{(i)}$, measured by a function $\mu(x^{(i)}, \hat{y}^{(i)})$. A lower semantic distance indicates more leakage, since the linked anonymised document is more similar to the original document. Concretely μ uses an LLM to compare each claim in the original document against the linked anonymised one, rating how far it is supported there. Only claims that remain supported will therefore contribute towards leakage.

Putting all these parts together, privacy is defined as the average semantic distance from applying both these stages across the entire dataset:

$$\text{semantic distance } (\uparrow) = \mathbb{E}_{x^{(i)} \in \mathcal{D}_{\text{original}}} [\mu(x^{(i)}, L(\tilde{x}^{(i)}, \mathcal{D}_{\text{anonymised}}))],$$

For improved clarity, we rename their privacy measure from **Semantic Distance** to **Semantic Leakage**, and invert it, and scale between 0 and 100. This means a higher Semantic Leakage corresponds to an attacker that can recover more correct information from the anonymised dataset.

$$\text{semantic leakage } (\downarrow) = 100 - 100 \times \text{semantic distance}$$

Using this framework may invite the objection that we anonymise against a linkage adversary and are evaluated by one. We therefore set out how the two adversaries differ.

The anonymisation and evaluation adversaries differ in three main ways. Firstly, **scope**. Semantic Leakage is a dataset-level quantity since the linker uses auxiliary information to select from every anonymised record, whereas our adversaries try to match auxiliary information to the single document it was derived from. Semantic Leakage is therefore not a property of a document in isolation, which is what the adversaries in our framework measure, and so cannot be optimised for directly by an anonymiser that runs independently per document. Secondly, **linkage**. Evaluation uses a dense retriever whereas our adversaries are language models that return confidence scores, a different class of model scoring by a different mechanism. Thirdly, **auxiliary information**. At evaluation this is three facts per individual, sampled independently of our anonymisation, whereas we partition an entire document’s facts across U adversaries. The particular conjunction used at evaluation is therefore, in general, not one of the subsets held by any adversary during anonymisation.

More importantly, our privacy gains are corroborated under a different threat model that we do not optimise for. On DB-BIO we report a success rate and confidence score superior to RUPTA, despite this being measured under an attribute inference threat model: success rate falls $65.27\% \rightarrow 47.70\%/53.56\%$ and confidence score falls $92.21 \rightarrow 79.30/64.21$. Specifically, success rate is measured by selecting from a list of 10 candidate names. Unlike us, RUPTA directly optimises for this metric, scoring privacy during anonymisation by generating an ordered list of the most likely celebrity names and ranking privacy by how far down the list the correct name is. RUPTA is therefore anonymising against very nearly the same measure it is later evaluated on for success rate and still performs worse than both of our methods.

3.5.2 Utility

We evaluate utility using both downstream performance and more general measures.

On DB-BIO we use the downstream task proposed by Yang et al. [20] of occupation classification. We use the same classifier they trained to run on our results, reporting both accuracy and loss.

For MEDQA, consistent with previous work, we evaluate utility using the accuracy of an LLM in guessing the correct answer using the anonymised patient description [17, 19].

As we described in the Section 2.4.2, we wish our anonymised texts to be similar in form to the original ones. As Vinod et al. [18] note, evaluating utility based on downstream performance alone could allow poor-quality texts to “hack” the evaluation metrics by inserting relevant keywords for the task. This is a risk for RUPTA on DB-BIO, since occupation labels are given during anonymisation and are later used as the ground truth for the downstream task evaluation.

We therefore use MAUVE, which measures how closely open-ended machine-generated texts are to human-written text by comparing the distribution of their embeddings [14]. We adopt MAUVE as a measure to quantify how plausibly our anonymised texts could have been drawn from the distribution of the original ones.

Since it was introduced at NeurIPS 2021, where it received an Outstanding Paper Award, it has been widely adopted for evaluating open-ended text generation. Unlike classical measures of a textual utility like BLEU or ROUGE, that use n-gram overlap, MAUVE evaluates textual quality semantically and at the dataset level. The authors show that it has strong correlations with human judgments.

Finally we use an LLM judge of readability of the texts. Staab et al. [17] introduce this as a measure of utility, and find alignment between it and a human annotated scores.

3.5.2.1 Accuracy $\uparrow$ (DB-BIO only)

Accuracy of an occupation classifier trained on original documents and tested on held-out anonymised documents. We include this measure because it replicates the evaluation setup used by RUPTA [20], allowing direct comparison.

3.5.2.2 Loss $\downarrow$ (DB-BIO only)

Cross-entropy loss of the same occupation classifier. This captures classifier confidence, not just correctness: a method can maintain accuracy while increasing loss, indicating less robust predictions.

3.5.2.3 Question Accuracy ↑ (MEDQA only)

Accuracy of an LLM judge in selecting the correct multiple-choice answer using the anonymised patient vignette. This tests whether anonymised documents preserve enough correct clinical information to support medical reasoning on professional board-level questions.

3.5.2.4 MAUVE ↑ (DB-BIO and MEDQA)

Distributional similarity between the set of original documents and the set of anonymised documents, measured between 0–1 [14]. This measures whether anonymised documents remain representative of the original dataset: a method can preserve per-document utility while shifting the overall distribution, creating an out-of-distribution problem for downstream training.

3.5.2.5 Readability ↑ (DB-BIO and MEDQA)

LLM-judged readability on a 1–5 Likert scale (1 = unreadable, 5 = excellent), reported as a percentage of the original document’s score to allow comparison across datasets. This measures whether anonymisation degrades text quality independently of factual correctness. Staab et al. [17] use this measurement for utility and validate the LLM-judged readability against human-annotated scores.

Chapter 4

Results

Our aim is to anonymise text against an adversary with auxiliary information about a target, rather than one who infers PII, whilst preserving both the general and task-specific utility of the anonymised document, and without conditioning on a downstream task.

4.1 Set Up

For our main privacy metric, semantic Leakage (see Section 3.5.1.3), we select the semantic retriever, Contriever [6] for the linkage function, and gpt-4o-2024-08-06 as the model for evaluating semantic distance. Readability is assessed using gpt-5.4-mini-2026-03-17 as judge.

For MAUVE, the authors emphasise that scores are meaningful for relative comparisons and not absolute scores. We therefore only discuss it for anonymisation methods applied to the same dataset. Following the authors guidance on adjusting the scaling factor for smaller clustered values, we use a factor of 1. Since this only affects absolute values and not relative ordering, this adjustment does not change any of the comparisons that we draw.

We ran small, anonymised experiments with differing values of privacy threshold τ , and number of adversaries U , however due to budget and time requirements, we provide no systematic sweep of these values. We find that anonymisation works better using lower values of τ , for ALTA compared to ALTA-FD. We believe the adversaries in ALTA are more confident since generalisation of the document never falsifies their background information unlike ALTA-FD which perturbs the document.

For details of our dataset specific set up, for DB-BIO see Section A.1, and for

MEDQA see Section A.2.

4.2 Experiment 1 - DB-BIO

Table 4.1 reports our results on DB-BIO. ALTA and ALTA-FD drastically improve privacy both under reconstruction attack (Semantic Leakage) and under RUPTA’s original privacy metrics of Success Rate and Confidence Score, whilst also improving the general utility metrics of readability and MAUVE. In addition, ALTA-FD achieves comparable task accuracy, despite the dataset-specific advantage given to RUPTA of access to the label which is later used for evaluation.

	Metric	Original	RUPTA	ALTA	ALTA-FD
Privacy	Semantic Leakage ↓	99.94	43.73	<u>9.26</u>	6.41
	Success Rate ↓	100.00%	65.27%	47.70%	<u>53.56%</u>
	Confidence Score ↓	99.55	92.21	<u>79.30</u>	64.21
Utility	Accuracy ↑	99.58%	98.33%	88.70%	<u>97.07%</u>
	Loss ↓	0.0422	0.0784	0.4823	<u>0.1632</u>
	MAUVE ↑	1.0000	0.5872	<u>0.6311</u>	0.9630
	Readability ↑	100% (3.39)	86.54% (3.07)	<u>92.11% (3.13)</u>	96.02% (3.26)

Table 4.1: Results from 239 anonymised documents. First place per metric, when comparing anonymisation methods, is shown in bold; second place is underlined.

RQ1 Does our proposed adversarial model improve anonymisation privacy?

Both ALTA and ALTA-FD show noteworthy improvements in privacy, with semantic leakage drastically reducing from RUPTA 43.73 $\rightarrow$ 9.26/6.41. This privacy result is corroborated, by our other privacy measurements with reductions in success rate 65.27 $\rightarrow$ 47.70/53.56 and confidence score 92.21 $\rightarrow$ 79.30/64.21 for both ALTA and ALTA-FD respectively.

We believe the noteworthy reductions in semantic leakage, moving from RUPTA’s to ALTA’s is indicative of the improvements of our adversarial method of matching auxiliary information with an ensemble of adversaries over entity generalisation. These results are corroborated by our other privacy measures of success rate and confidence score, despite being better aligned with the threat model assumed by RUPTA during anonymisation.

RQ(2) Can fact distortions more readily preserve document style and readability?

We isolate the effects of fact distortion directly by comparing ALTA to ALTA-FD since

Henry Benjamin "Hank" Greenberg (born Hyman Greenberg; January 1, 1911 – September 4, 1986), nicknamed "Hammerin' Hank", "Hankus Pankus" or "The Hebrew Hammer", was an American professional baseball player. He played in Major League Baseball (MLB), primarily for the Detroit Tigers as a first baseman in the 1930s and 1940s. A member of the Baseball Hall of Fame, he was one of the premier power hitters of his generation and is widely considered as one of the greatest sluggers in baseball history.	An individual, recognized by several aliases, was a professional athlete. This person participated in a prestigious sports league, mainly for a primary organization, during a prior era. A member of a distinguished sports institution, they were among the top performers of their time and are broadly acknowledged as one of the finest in sports history.	Nathaniel J. Turner (February 9, 1908 – April 13, 1989) was an American professional baseball player. He played in Major League Baseball primarily as a third baseman in the 1930s and 1940s. Turner played for a team based in the Central United States and spent most of his career on this team, though he was later transferred to the Boston Red Sox. His career was marked by his excellent defensive skills and he is recognized among notable baseball players.
Original	RUPTA	ALTA-FD

Figure 4.1: Example of how anonymisation under ALTA-FD can improve readability and biographical style compared to RUPTA

this is a clean ablation. Performance improves on six of the seven metrics: Semantic Leakage $9.26 \rightarrow 6.41$, Confidence $79.30 \rightarrow 64.21$, Accuracy $88.70 \rightarrow 97.07$, Loss $0.4823 \rightarrow 0.1632$, MAUVE $0.6311 \rightarrow 0.9630$, Readability $92.11 \rightarrow 96.02$. Only Success Rate worsens $47.70 \rightarrow 53.56$. Notably this includes improvements on all four utility metrics, with improvements occurring on both general and task specific metrics.

When comparing the results of RUPTA to ALTA-FD we find better biographical plausibility (MAUVE: $0.5872 \rightarrow 0.9630$), improved readability ($86.54\% \rightarrow 96.02\%$), and comparable task accuracy ($98.33 \rightarrow 97.07$). Some reduction in task utility from RUPTA is not unexpected given our task-agnostic approach and we see an increase in the loss ($0.0784 \rightarrow 0.1632$), meaning the classifier is less confident. We more closely examine document style and readability through an example in Figure 4.1.

ALTA-FD uses a pseudonym “*Nathaniel J Turner*”, perturbs the birth and death dates, the players position from a first baseman to be a third baseman, and invents that the player played for the Boston Red Sox. Although these details are untrue, the same general trajectory of a notable American Major league Baseball player, who played in the 1930s-1940s is preserved. Notably, the text reads like a natural, genuine biography.

Conversely, RUPTA generalises most of the named entities in the document. The anonymisation describes “*an individual*”, that is a “*professional athlete*” for a “*primary organisation*”, in a “*Prestigious sports league*”. Although all of these are true, this does not reconcile both the lost information (the sport, the league, the team, the time-period) and the reduced readability. Additionally, this no longer reads like a natural biography would but like a redacted one.

RQ(3) To what extent does our task-agnostic approach reduce task-specific utility?

Task utility falls for both ALTA and ALTA-FD from the unanonymised documents, though for different reasons. ALTA’s larger accuracy drop is likely from genuine utility reduction from coarsening or removing the occupation from the anonymised document. This is confirmed by our ablation in RQ(2), where the addition of perturbation recovers accuracy from $88.70\% \rightarrow 97.07\%$ and loss from $0.4823 \rightarrow 0.1632$, whilst maintaining similar levels of privacy.

ALTA-FD’s task accuracy remains comparable to RUPTA, but loss increases, indicating the classifier is less confident in its predictions of occupation: loss increases from 0.0784→0.1632 but accuracy only falls 98.33→97.07. This is indicative of a familiar signal in an unexpected form rather than an absent one. This is partly expected since RUPTA’s utility-mode rewriting conditions on the ground-truth labels, prompting utility rewriting with

“You should now improve the utility score about classifying the text as {label} without decreasing the privacy score.”

Yang et al. [20] note that since the classifier is trained on unanonymised documents, it creates an out-of-distribution problem for evaluation. They attribute their own small degradations in utility to closer preservation of the original text. A fine-tuned BERT-classifier is sensitive to changes in the lexical surroundings it was fine-tuned on and so classification performance measures both lexical similarity to the original document alongside actually preserving signals to guessing the occupation.

RUPTA’s rewriting prompt prioritises minimal word changes and generalises entities, whereas ALTA-FD reconstructs the anonymised document using its distorted fact set. When comparing lexical similarity of the anonymised texts to the original, RUPTA is higher, preserving more of the original documents tokens, which we confirm through measuring Jaccard Similarity and Token Overlap: RUPTA (Jaccard Similarity: 0.373, Token Overlap: 0.421), ALTA-FD (0.256, 0.318), ALTA (0.155, 0.159). Therefore, for similar task accuracy, it is difficult to discern whether RUPTA’s improvements in loss come from greater lexical similarity to the original document or from genuinely better preserving clues to the celebrities occupation.

This confounding factor applies equally to our own ablation of ALTA to ALTA-FD, where increased task-specific utility could be attributable to a higher lexical overlap with the original document (Jaccard Similarity: 0.155 vs 0.256), rather than better preservation of occupation signal. However, our MEDQA results for Section 4.3 also show improvements from ALTA to ALTA-FD for task specific utility (70.4% → 72.4%). Evaluation there is done zero-shot by an LLM rather than a fine-tuned classifier, and so is not confounded by lexical overlap, which suggests the gains from fact distortion are genuine.

ALTA-FD remains within 1.3 percentage points of RUPTA on classification accuracy despite rewriting from its distorted fact set and without ever seeing or optimising for the task label, on a metric that favours lexical similarity. We therefore believe that even though the loss is higher, task specific utility is still largely preserved under ALTA-FD.

4.3 Experiment 2 - MEDQA

ALTA and ALTA-FD improve on privacy and all utility metrics over AF. Both task specific utility and general utility improves, with document plausibility increasing for both ALTA and ALTA-FD. Across all methods, ALTA-FD achieves both the best privacy and question accuracy. We find this result notable given how sensitive correct diagnosis is to information such as symptoms or patient background, that perturbation could easily destroy.

	Metric	Original	Adversarial Feedback	ALTA	ALTA-FD
Privacy	Semantic Leakage ↓	91.23	25.79	<u>24.74</u>	21.48
	Question Accuracy ↑	88.0%	66.8%	<u>70.4%</u>	72.4%
Utility	MAUVE ↑	1.000	0.748	<u>0.903</u>	0.940
	Readability ↑	100% (4.15)	<u>83.8% (3.64)</u>	89.7% (3.83)	<u>83.8% (3.64)</u>

Table 4.2: $N = 250$, run on a randomly sampled subset of MEDQA. First place, when comparing anonymisation methods, is shown in bold; second place is underlined.

RQ1 Does our proposed adversarial model improve anonymisation privacy?

We find reductions in semantic leakage from AF from $25.79 \rightarrow 24.74/21.48$ for ALTA and ALTA-FD respectively. Although these privacy gains are modest, this is whilst beating AF on all utility metrics.

RQ(2) Can fact distortions more readily preserve document style and readability?

When comparing the results of AF to ALTA-FD we find better medical profile plausibility (MAUVE: $0.748 \rightarrow 0.950$), equal readability (83.8%), and improved task accuracy ($66.8\% \rightarrow 72.4\%$). Under AF explicit PII information related to the patient is removed, which frequently occurs in MEDQA in its surface form. This means AF cannot create anonymised documents that read similar to the original.

We isolate the effect of fact distortion by comparing ALTA to ALTA-FD since this is a clean ablation. Performance improves on three of the four metrics: Semantic Leakage $24.74 \rightarrow 21.47$, Question accuracy $70.4\% \rightarrow 72.4\%$, MAUVE $0.903 \rightarrow 0.940$. Readability reduces slightly from $89.7\% \rightarrow 83.8\%$.

We more closely examine document style and readability compared to AF through an example in Figure 4.2. Both examples evade re-identification under our privacy reconstruction attack.

AF removes any surface level mentions of PII. The patients age, “6 year old”, nationality “African American” and gender “boy” are all compressed into “An individ-

ual”. Symptoms are also heavily generalised since sickle cell is more common among individuals with African ethnicity. Diagnostic test results are also completely removed from the anonymisation. The anonymised document that remains reads much less like a patient profile, but a heavily redacted one.

ALTA-FD keeps PII at its surface form but perturbs it. “6 year old” → “11 year old”, “African American” → “Middle Eastern”. Unlike AF, not only information related to PII is anonymised but any information that could allow re-identification. “treated with over the counter analgesics” → “treated with herbal remedies”, “pain in his hands” → “pain in his shoulders”. Instead of removal, lab results are perturbed to different, but still elevated levels. Importantly, these changes allow the anonymised document to read much more like a patient profile, whilst maintaining the correct inferences of sickle cell.

A 6-year-old African American boy is referred to the hospital by his family physician for jaundice, normocytic anemia, and severe bone pain. He has a history of several episodes of mild bone pain in the past treated with over the counter analgesics. On physical examination, the child is icteric with nonspecific pain in his hands ... Total bilirubin 8.4 mg/dL, WBC 9,900/mm³, Hemoglobin 6.5 g/dL, MCV 82.3 fL, Platelet count 455,000/mm³ ...	An individual is referred to a healthcare facility for a medical condition involving discomfort and pain. The individual has a history of similar episodes in the past treated with common pain relievers. On examination, the individual shows signs of discomfort with nonspecific pain in their extremities ... A series of diagnostic tests are performed:	An 11-year-old Middle Eastern boy was referred to the hospital by a medical professional. He exhibits yellowing of the eyes and has normocytic anemia. In addition to severe bone pain, he has a history of bone pain and was treated with herbal remedies. His skin appears pale and he experiences nonspecific pain in the shoulders ... total bilirubin of 13.0 mg/dL, a white blood cell count of 14,000/mm³, a hemoglobin level of 10.4 g/dL, mean corpuscular volume of 64.7 fL, and a platelet count of 680,000/mm³ ...
Original	AF	ALTA-FD

Figure 4.2: Anonymised medical vignette under AF and ALTA-FD. Both prevent re-identification and the correct answer is obtained, but under ALTA-FD the anonymised document that reads like a medical profile, whereas under AF it does not. Notably all diagnostic test results are removed under AF and ends abruptly, whereas the test values are perturbed to other elevated values under ALTA-FD.

RQ(3) To what extent does our task-agnostic approach reduce task-specific utility?

There is a meaningful reduction in task-specific utility shown by question accuracy reducing across all anonymisation methods. We compare to the upper bound given by the original patient vignettes (Question Accuracy 88%), where both ALTA and ALTA-FD show less reductions in accuracy (88% → 70.4%/72.4%) over AF (88% → 66.8%)

ALTA-FD, which obtain the highest question accuracy, perturbs information in the document, adding unsupported synthetic information. Given the difficulty of the questions (professional medical board examination questions), and in a domain as precise as medicine, we believe this result is particularly noteworthy. Indeed, minor perturbations of symptoms or background could completely change the correct inferable diagnosis.

When comparing ALTA to ALTA-FD, which isolates the addition of perturbation operations, we see an increase in task accuracy from 70.4% to 72.4%. For question accuracy to increase would imply that more correct medical inferences are supported. Therefore, fact level rewriting to include perturbation operations, allows greater recovery

of correct medical inferences than simple generalisation. This also supports that the perturbations made by ALTA-FD are more useful than destructive (since both task accuracy and privacy increase from ALTA). We provide an example to demonstrate how a good perturbation can allow inferences not possible under generalisation in Section 2.4.2.

PII is sometimes fundamentally tied to the content of a document. For MEDQA, some questions centre on patients with Alzheimer’s (age) or with Pregnancy (sex). AF tries to remove inferences to these PII identifiers, yet their inference may be necessary to correctly answer the questions. ALTA on the other hand, is more flexible in its anonymisation, only generalising information where it is deemed highly identifying. We believe this as an explanation for why under ALTA’s adversarial we see improvements in question accuracy $66.8\% \rightarrow 70.4\%$, whilst privacy also improves.

4.4 Discussion

4.4.1 Summary

ALTA and ALTA-FD obtain their privacy by different mechanisms. We remind the reader from Section 3.5.1.3 that the attack happens in two stages: linkage, where the adversary picks the anonymised record, it believes is its target’s, and leakage, where the true information recoverable from that record is measured. We define the linkage rate to be the proportion of individuals whose auxiliary information is linked to their own anonymised record, which is shown by the horizontal edges in Figure 4.3. We note that the linkage rate is merely diagnostic and not a privacy metric itself but allows us to understand where in the simulated attack privacy is compromised. On DB-BIO, the linkage rate drops from 70.3% under RUPTA, to 63.6% under ALTA-FD and to 34.7% under ALTA. The same direction between ALTA-FD and ALTA holds on MEDQA, though the gap between the two methods is far smaller. It therefore seems generalisation-only rewriting is more successful at breaking linkage, since a generalised document will plausibly match more adversaries’ background information, whereas fact distortion permits linkage but contaminates what can be learned from the linked document through introducing synthetic information.

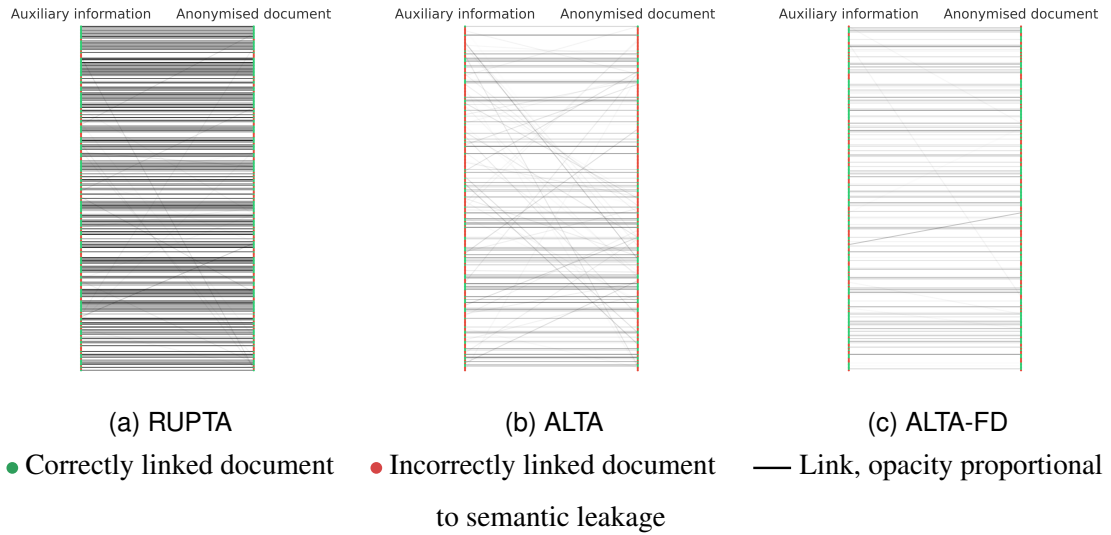


Figure 4.3: Visualisation of the reconstruction attack used for the **Semantic Leakage** metric, for RUPTA, ALTA and ALTA-FD. Each small node (more visible when zoomed in) represents an individual. The left column holds the **auxiliary information** the adversary has about each individual, and the right column holds the corresponding **anonymised documents**. A individual appears at the same height in both columns, so a horizontal edge indicates that the adversary linked an individual’s auxiliary information to the correct anonymised document. Nodes are green where this link is correct and red where it is not, and each edge is drawn in black with opacity proportional to the semantic leakage from the document it leads to. ALTA and ALTA-FD leak substantially less than RUPTA, visible in both the fewer horizontal edges and their lower opacity.

4.4.2 Limitations

Perturbations can Produce Contradictions or Verifiably False Information. Any synthetic method can produce information that is contradictory or verifiably false. ALTA-FD inherits this limitation by including a perturbation operation. For DB-BIO, it is easy for named entities such as places, people and dates to no longer align with publicly verifiable records; the example in Figure 4.1 invents a transfer to the Boston Red Sox, which is verifiable against public records. We stress test the impact of such contradictions on task utility by choosing MEDQA, where contradictory clinical details could produce diagnostic uncertainty and erroneous diagnosis. Despite this, we find improvements on question accuracy from a pure generalisation approach of ALTA $70.4\% \rightarrow 72.4\%$ whilst improving privacy $24.74 \rightarrow 21.48$, indicating the harm of any contradictions is outweighed by the retention of information removed by pure generalisation. We would caution against production use of a synthetic method like ALTA-FD,

for datasets where information is easily publicly verifiable, given the difficulty in forging correct synthetic information, or for heavily technical domains, where determining a non-contradictory replacement is difficult.

Adversary Auxiliary Information is Randomised and Fixed Throughout Anonymisation. All anonymisation methods must make an assumption about what auxiliary information is available to an adversary. In ALTA and ALTA-FD, adversaries within the anonymisation framework are supplied with randomly selected facts from the original document as auxiliary information. This auxiliary information remains fixed across each iteration for a given document, and we anonymise against only the feedback from the U adversaries relating to their **specific** background information. Since we do not enumerate every possible subset of auxiliary information, our anonymisation targets these specific adversaries as a proxy and not every adversary with arbitrary background information. Auxiliary information held by a real adversary could be both more likely knowable and more identifying than the random subsets we anonymise against.

Auxiliary Information is Not only Document Derivable. We assume all auxiliary information known by the adversaries is derivable from the document, when in reality, information outside of it could be known. For example, a text about someone sleeping in the cold and with a few coins, does not tell us they are homeless, but if known to an adversary, would help in re-identification. This is a limitation of any anonymisation method unless further information is known and used by it. Our approach has a natural extension if such information is known: append extra known information to our document derived facts and distribute them between our U adversaries.

4.4.3 Future Work

We direct future work to continue the development of adversarial anonymisation methods that use auxiliary linkage rather than inference of PII. We believe the results on MEDQA of ALTA-FD demonstrate that perturbations can be successfully incorporated into textual anonymisation for challenging downstream task. Greater exploration of using perturbation for anonymisation of text would be beneficial. Finally, the mechanism of grouping facts to form auxiliary information should be explored beyond random, for example, by assessing the plausibility of facts being reasonably known individually and together, or by constructing the most identifying auxiliary information at a fixed cardinality, as a worst case to anonymise against.

Chapter 5

Conclusions

Anonymisation of text is dominated by methods that use Personally Identifiable Information (PII) as the unit of protection. Industry adopted approaches detect and remove surface level PII whereas recent LLM-adversarial frameworks use inferable PII. However, this does not protect against genuine adversaries who may have auxiliary information beyond PII, that still permit re-identification. Data protection legislation centres upon outcome, and whether an individual can be re-identified by the means “reasonably likely” to be used, which is independent of whether a finite list of PII identifiers is removed.

In this dissertation, we argue for changing the adversary central to these LLM-adversarial frameworks from inferred PII to linkage of auxiliary information. We design an anonymisation method, ALTA, that uses an ensemble of adversaries each with auxiliary information that they try to link to the anonymised document. ALTA-FD builds upon ALTA with a fine-grained rewriting process, through decomposing the text into a list of facts, transforming them through generalisation and perturbation, and reconstructing them into a document. Both these methods attempt to preserve utility of the anonymised text, but unlike previous work, do so in a general task-agnostic way, to support better data reusability.

Across two datasets, both methods improve privacy beyond existing state-of-the-art adversarial LLM-based methods. On DB-BIO semantic leakage falls from 43.73 under RUPTA to 9.26/6.41 for ALTA and ALTA-FD, a reduction of over 85%, and corroborated by reductions in success rate and adversary confidence. On MEDQA, ALTA and ALTA-FD improve upon privacy while matching or improving on every utility measure against Adversarial Feedback. General utility improves across both datasets for both methods: MAUVE rising from 0.587 to 0.963 on DB-BIO and from

0.748 to 0.940 on MEDQA, with qualitative examples illustrating how anonymised documents read more as the same kind of style as the original rather than redacted ones.

Task-specific utility is a particular challenge for our methods after opting for a task-agnostic approach. On MEDQA, question accuracy improves over the baseline, from 66.8% to 70.4% and 72.4%. On DB-BIO ALTA-FD remains within 1.26 percentage points of a task-specific baseline that receives the ground-truth label during anonymisation, and which is evaluated by a classifier that rewards lexical similarity to the original. We regard the MEDQA result as more informative since task-utility is unconfounded by lexical similarity to the original, and the task of answering professional medical board examinations more challenging. In particular, it tests the limits of ALTA-FD since perturbing a clinical detail could change the correct answer. Despite this, ALTA-FD improves both privacy and task accuracy, and is the clearest evidence that perturbation, when applied carefully at the level of individual facts, can recover information that is destroyed under pure generalisation.

A secondary important finding is how privacy is obtained for both methods. ALTA which uses generalisation, reduces the adversary's ability to link its auxiliary information to the correct document, whereas the introduction of perturbation in ALTA-FD more readily permits linkage, but contaminates the information recoverable from it. This complements the account given in Chapter 2 and strengthens the case for combining the two operations over applying them separately.

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Appendix A

First appendix

A.1 DB-BIO Set Up

We run RUPTA with default settings used in their paper, matching ALTA and ALTA-FD for the number of anonymisation iterations $T = 5$ and LLM model of gpt-4o-2024-08-06.

We evaluate success rate and confidence score using GPT-5.4-2026-03-05. This simulates a stronger adversary, with better reasoning and knowledge than that used for anonymisation.

We set privacy threshold, τ , to 50 and 70 for ALTA and ALTA-FD respectively.

A.2 MEDQA Set Up

We run Adversarial Feedback (AF), anonymising over the full set of personal identifiers provided, consistent with previous work applying it to MEDQA [19]. We use $T = 5$ iterations across all models and LLM model gpt-4o-2024-08-06 for anonymisation.

We evaluate question accuracy using GPT-5.4-2026-03-05, applying it zero shot to select the correct multiple choice answer using the anonymised patient profile, the original question and multiple choice options. This simulates a test taker with greater domain knowledge and reasoning than the model used for anonymisation. Since task-utility is evaluated zero-shot and no classifier is trained, it is not confounded by lexical similarity to the original patient profile unlike our earlier measurements of task-utility on DB-BIO.

We use smaller values of τ , reflecting the sensitive nature of the downstream task, of $\tau = 40$ and $\tau = 60$ for ALTA and ALTA-FD respectively. This maintains the same

difference of 20 between them we use when applying our methods to DB-BIO.

A.3 Inferrable PII

We give an example of *inferrable PII*, which is PII not expressed explicitly, but can be inferred using context-dependent clues, by using an example and explanation from [17]:

This left shark thing is hilarious, I've seen it, walked just by there to see it live after my final exams.

No locations are stated here, but the *left shark incident* is in reference to a 2015 Superbowl show at half time. This allows the inference of the PII attribute location as being Glendale without Glendale ever explicitly being stated in the text.

A.4 Formal Explanation

A.4.1 Setup

A.4.1.1 Iterative Rewriting

Let x_{orig} represent some original document, where x_t represents the rewritten document at iteration t and $x_0 = x_{\text{orig}}$.

We iteratively rewrite the document for T iterations or exit earlier if the current documents privacy score exceeds τ and utility score exceeds 95.

To produce the next document, x_{t+1} , we use the adversarial feedback fb_t and document x_t for the current version.

ALTA writes x_{t+1} directly using x_t and fb_t , whereas ALTA-FD does so in several stages through a process we term *fact distortion*: iteratively transforming facts and rewriting the document from these transformed facts each iteration.

A.4.1.2 Fact Decomposition

Let the function F decompose the original document x_{orig} into a set of atomic facts:

$$F_{\text{orig}} := F(x_{\text{orig}}) = \{f_1, f_2, f_3, \dots, f_n\} \quad (\text{A.1})$$

where each f_i is an atomic fact and the document is decomposed into n total facts.

A.4.2 Adversarial Feedback

A.4.2.1 Auxiliary Linking

Randomly partition the document's facts F_{orig} into U disjoint subsets of equal size $A_1, \dots, A_U$. Formally,

$$F_{\text{orig}} = \bigsqcup_{j=1}^U A_j.$$

Each subset A_j represents the auxiliary information an adversary j has about the subject described in x_{orig} .

$$fb_t^j, c_t^j = M_j^{\text{adv}}(x_t, A_j) \quad (\text{A.2})$$

For each adversary j of the U adversaries, we prompt an LLM to assess how confident it is that the individual represented in its auxiliary information A_j is the same individual represented in the current document x_t . Let c_t^j denote the LLM's confidence in linking A_j to x_t , ranging from 0 to 100. We define our privacy heuristic score as:

$$p_t^j = 100 - c_t^j, \quad c_t^j \in [0, 100]. \quad (\text{A.3})$$

We also obtain feedback fb_t^j describing what information in x_t most enables adversary j to link its auxiliary information A_j .

A.4.2.2 Synthesised Feedback

Using the feedback $\{fb_t^1, \dots, fb_t^U\}$, we use an LLM to synthesise a summary of the feedback identifying which pieces of information in the text most confidently allow re-identification.

$$fb_t = M^{\text{syn}}\left(x_t, \bigcup_{j=1}^U fb_t^j\right) \quad (\text{A.4})$$

We let the privacy score equal a weighted sum of the minimum privacy score across adversaries and the mean. This allows the documents privacy to be reflective of privacy as a worst case scenario, an average case one, or somewhere in between.

$$p_t = \gamma \cdot \min_j(p_t^j) + (1 - \gamma) \cdot \text{mean}_j(p_t^j), \quad \gamma \in [0, 1] \quad (\text{A.5})$$

A.4.3 Iterative Rewriting

A.4.3.1 ALTA (Auxiliary Linking Textual Anonymisation)

Using the synthesised feedback, we instruct an LLM to anonymise the text so as to reduce re-identification cues present in the adversaries’ feedback:

$$x_{t+1} = M^{\text{anon}}(x_t, fb_t) \quad (\text{A.6})$$

The LLM is instructed to only rewrite by making information more general, and never to introduce false information into the document.

A.4.3.2 ALTA-FD (Auxiliary Linking Textual Anonymisation with Fact Distortion)

Instead of rewriting the document in one go, we introduce the idea of fact-level transformations. These are used to reduce the linkage confidence of adversaries by targeting facts in the anonymised document and changing them. We term this process of applying fact transformations *fact distortion*, and use the terms “transformation” and “distortion” interchangeably.

A.4.3.2.1 Transformations

- **Coarsen**: state a strictly more general version of the fact
(e.g. “a lightning-bolt scar on his forehead” → “a scar on his forehead”).
- **Perturb**: state a different but equally specific fact at the same level of abstraction
(e.g. “he liked cats” → “he liked dogs”).
- **Specialise**: keep the fact but add an invented refinement at a finer level of detail
(e.g. “he liked dogs” → “he liked poodles”).
- **Delete**: remove the fact entirely.
- **Identity**: leave the fact unchanged.

Each transformation has a strength: **Low**, **Medium**, or **High**.

Low: For coarsening, remove only minor qualifying detail or move one hypernym step. For perturbation, move to a sibling: a different member of the same narrow subcategory. For numerical facts, move to adjacent values within the same category. When to apply: facts that are similar to their original versions but are barely or not mentioned in the adversary’s feedback.

Medium: For coarsening, remove most qualifying details or move several taxonomic levels. For perturbation, move to a cousin: the same broad category but a different subcategory, region, or era. For numeric quantities retain the general direction or magnitude but make it more or less extreme.

When to apply: typical re-identifying facts noted by the adversary.

High: For coarsening, keep only the core category. For perturbation, move to a distant relative: the same type of thing, maximally different while remaining plausible. For numeric quantities, large changes that reach the boundaries of adjacent grouped values. Before choosing high, check what other facts must change with it to stay consistent.

When to apply: ONLY where a fact has already been transformed and continues to be re-identifying, OR where the adversary indicates very directly a large amount of re-identification risk.

A.4.3.2.2 Transformation Assignment To transform the facts, we first *assign* a transformation type and strength to each fact from the previous iteration:

$$TA_{t+1} = M^{TA}(F_0, F_t, fb_t, I) \quad (A.7)$$

where F_{orig} is the original document’s facts, F_t is the most recent set of distorted facts for the anonymised document x_t , fb_t is the synthesised adversarial feedback, and I is an optional argument holding conclusions and inferences drawn from the original document, so that transformations are picked such that certain general inferences can still be made from the anonymised document.

TA_{t+1} is a list of tuple pairs corresponding to the LLM-selected transformation type and strength for each fact:

$$TA_{t+1} = [(D_{t+1}^1, S_{t+1}^1), (D_{t+1}^2, S_{t+1}^2), \dots, (D_{t+1}^n, S_{t+1}^n)] \quad (A.8)$$

where D_{t+1}^i and S_{t+1}^i represent the distortion type and strength applied to fact i from document x_t , to produce the next rewriting x_{t+1} .

A.4.3.2.3 Transformation Rewriting To apply the fact transformations, we prompt a separate LLM to rewrite each fact according to a set of rules for its transformation type and strength:

$$F_{t+1} = M^{TR}(F_t, TA_{t+1}, fb_t, I) \quad (A.9)$$

Again, I is an optional argument to help fact rewriting to be more conducive of the original documents conclusions and inferences. This produces the new set facts F_{t+1} :

$$F_{t+1} = \{f_1^t, f_2^t, \dots, f_n^t\}. \quad (\text{A.10})$$

Note that

$$|F_{t+1}| \leq |F_t| \leq |F_{t-1}| \leq \dots \leq |F_{\text{orig}}|$$

due to the deletion transformation removing a fact from all future fact sets.¹

A.4.3.2.4 Document Rewriting After producing the distorted facts for the next iteration, F_{t+1} , they are used to write document x_{t+1} :

$$x_{t+1} = M^{\text{DR}}(F_{t+1}, x_0) \quad (\text{A.11})$$

For the rewriting, the facts in F_{t+1} are written into fluent prose, where the original written document x_0 is optionally provided to preserve its structure and phrasing. We believe including x_0 should produce documents more fluent, and aligned in style to the original. However, this does risk leaking information in the original that is no longer canonical in the distorted facts. We explicitly prompt the LLM that the distorted facts F_{t+1} are canonical, however if x_0 is provided, we can no longer guarantee that information outside of them will be included.

A.4.4 Utility Mode

As discussed in Section 3.2.3, we adopt the framing of anonymisation as optimisation of a lexicographic function of privacy and utility, where documents are judged first by their privacy and, when two documents are of equal privacy, by their utility.

We denote the privacy threshold τ as the level our heuristic privacy score, p_t , must exceed before our architecture switches to optimising utility. If the current anonymised document ever falls below τ again, we switch back to privacy mode.

Unlike RUPTA, which re-introduces utility using the ground-truth label, we assume a more general case in which the intended use of the anonymised document is **not known** prior to anonymisation.

¹This is with the exception of the utility mode, which can re-introduce deleted facts.

A.4.4.1 ALTA

If $p_t \geq \tau$, we optimise utility by replacing eq A.6 with:

$$x_{t+1} = M^{\text{util}}(x_t, fb_t, I) \quad (\text{A.12})$$

where x_t is the current document, fb_t is the privacy linkage feedback, and I is the inferences and conclusions made from the original document. The LLM is tasked with rewriting the document to better allow the inferences and conclusions to be drawn from x_t , whilst providing privacy feedback so that it weighs the privacy risks of these changes against the improvement in utility.

A.4.4.2 ALTA-FD

If $p_t \geq \tau$, we replace eq.. A.7 with:

$$TA_{t+1} = M^{\text{TA}}(F_0, F_t, fb_t, H, I) \quad (\text{A.13})$$

where F_{orig} is the original set of facts, F_t are the current facts, H is the fact history, and I is the set of conclusions and inferences from the original document.

The process of fact transformation provides a natural way of reintroducing utility by reversing previous transformations. Unlike rewriting the entire document, this allows the LLM to select only the information important for supporting the inferences and conclusions I by re-introducing it, and avoid reintroducing highly identifying information.

We define our *reversal transformations* as follows:

1. **Identity**: leave the fact as is.
2. **Undo**: move a fact back α positions in its fact history H .
3. **Restore**: recover a fact from $F_{\text{orig}} \setminus F_t$ (the deleted facts).

We write these assigned fact reversals as

$$TA_{t+1} = [(R_{t+1}^1, \alpha_{t+1}^1), (R_{t+1}^2, \alpha_{t+1}^2), \dots, (R_{t+1}^n, \alpha_{t+1}^n)] \quad (\text{A.14})$$

a list of tuples in which each R_{t+1}^i is an assigned reversal transformation and α_{t+1}^i is the number of places to reverse for the “Undo” reversal transformation, in contrast to eq. A.8.

These fact reversals are applied mechanically in a deterministic way since they use existing facts in the fact history. The following equation therefore replaces eq. A.9, where G is a function that applies the reversal transformations:

$$F_{t+1} = G(TA_{t+1}, H). \quad (\text{A.15})$$

F_{t+1} is then used as usual in eq. A.11 to produce x_{t+1} .

A.5 Pseudocode

Here we present the pseudocode for both our proposed methods. We represent different type of calls to an LLM using M , with a superscript to indicate its meaning. Conceptually these can be thought of calls to different LLMs or as different models.

A.5.0.1 EVAL PRIVACY

1. M_j^{adv} - **Adversarial** model with auxiliary information A_j , applied as a judge to score how confidently it can link this to the anonymised document.
2. M^{syn} - **Synthesises** individual adversarial feedback into a single directive on what information in the document is most re-identifying.

A.5.0.2 EVAL UTILITY

1. M^{IS} - Applied as a judge to score how much the **inferences**, I , are **supported** in the anonymised document.

A.5.0.3 ALTA

1. M^{anon} - Using adversarial feedback, better **anonymises** the document, by making it more general.
2. M^{util} - Reintroduces **utility** by rewriting the document to better support inferences and conclusions derived from the original.

A.5.0.4 ALTA-FD

1. M^{TA} (Prompt ??) - **Transformation assignment** to be applied to the current documents facts.

2. M^{TR} - **Transformation Rewriting**, using the assigned transformations and applied to the current documents facts.
3. M^{DR} - **Document rewriting**, using the distorted facts and optionally the original document for structure.

A.5.0.5 Shared

1. M^{IC} - Extracts general **inferences and conclusions** from the original document, that should still be supported in the anonymised one.

Algorithm 1 EVAL_PRIVACY

Evaluates privacy by asking U adversaries, to state their confidence that the document describes their subject. Feedback from all adversaries is synthesised, and privacy is scored as a weighted sum of the minimum and mean privacy from each adversary.

Input: document x ; auxiliary information sets $\{A_1, \dots, A_U\}$

Output: synthesised adversary feedback fb^{syn} ; combined privacy score p

```

1:  $\gamma \leftarrow 0.8$ 
2: for  $j = 1$  to  $U$  do                                     ▷ adversary feedback loop
3:    $(fb^j, C^j) \leftarrow M_j^{adv}(x, A_j)$ 
4:    $p^j \leftarrow 100 - C^j$ 
5: end for
6:  $fb^{syn} \leftarrow M^{syn}(x, fb^1, \dots, fb^U)$                  ▷ synthesised feedback
7:  $p \leftarrow \gamma \cdot \min_j(p^j) + (1 - \gamma) \cdot \text{mean}_j(p^j)$ 
8: return  $(fb^{syn}, p)$ 
```

Algorithm 2 EVAL_UTILITY

Document utility is a weighted sum of an LLM-judged score for whether the original document's inferences remain supported, and cosine similarity to the original.

Input: document x ; original document x_0 ; inferences I

Output: utility score u

```

1:  $u \leftarrow 0.9 \cdot M^{IS}(x, I) + 0.1 \cdot \text{COSSIM}(x_0, x)$ 
2: return  $u$ 
```

Algorithm 3 BEST_TEXT

Selects the best rewritten text lexicographically, first prioritising privacy and then utility.

Input: document rewrites with scores $((x_0, p_0, u_0), \dots, (x_T, p_T, u_T))$; privacy threshold τ

Output: best anonymised document

```

1:  $sat\_priv \leftarrow \arg \max_t [\max(p_t, \tau)]$ 
2: if  $|sat\_priv| == 1$  then
3:   return  $x_{sat\_priv}$ 
4: end if
5:  $best\_util \leftarrow \arg \max_{t \in sat\_priv} u_t$ 
6: return  $x_{best\_util}$ 

```

Algorithm 4 ALTA

Input: original document x_{orig} ; max iterations T ; privacy threshold τ

Output: anonymised document

```

1:  $x_0 \leftarrow x_{orig}$ 
2:  $I \leftarrow M^{IC}(x_0)$  ▷ inferences and conclusions
3:  $F_{orig} \leftarrow F(x_0)$  ▷ fact decomposition
4:  $\{A_1, \dots, A_U\} \leftarrow \text{SPLITEQUALLY}(F_{orig})$ 
5: for  $t = 0$  to  $T$  do
6:    $(fb_t, p_t) \leftarrow \text{EVAL\_PRIVACY}(x_t, \{A_1, \dots, A_U\})$ 
7:    $u_t \leftarrow \text{EVAL\_UTILITY}(x_t, x_0, I)$ 
8:   if  $t = T$  then break
9:   end if
10:  if  $p_t \geq \tau$  and  $u_t \geq 95$  then
11:    return  $x_t$  ▷ early stopping
12:  end if
13:  if  $p_t < \tau$  then ▷ privacy mode rewrite
14:     $x_{t+1} \leftarrow M^{anon}(x_t, fb_t)$ 
15:  else ▷ utility mode rewrite
16:     $x_{t+1} \leftarrow M^{util}(x_t, fb_t, I)$ 
17:  end if
18: end for
19: return BEST_TEXT $((x_0, p_0, u_0), \dots, (x_T, p_T, u_T), \tau)$ 

```

Algorithm 5 ALTA-FD**Input:** original document x_{orig} ; max iterations T ; privacy threshold τ **Output:** anonymised document

```

1:  $x_0 \leftarrow x_{\text{orig}}$ 
2:  $I \leftarrow M^{\text{IC}}(x_0)$ 
3:  $F_0 \leftarrow F(x_0)$ 
4:  $\{A_1, \dots, A_U\} \leftarrow \text{SPLITEQUALLY}(F_0)$ 
5:  $H \leftarrow [((\text{original}, \text{None}), f_0^i) : i = 1, \dots, |F_0|]$  ▷ fact history
6: for  $t = 0$  to  $T$  do
7:    $(fb_t, p_t) \leftarrow \text{EVAL\_PRIVACY}(x_t, \{A_1, \dots, A_U\})$ 
8:    $u_t \leftarrow \text{EVAL\_UTILITY}(x_t, x_0, I)$ 
9:   if  $t = T$  then break
10:  end if
11:  if  $p_t \geq \tau$  and  $u_t \geq 95$  then
12:    return  $x_t$  ▷ early stopping
13:  end if
14:  if  $p_t < \tau$  then ▷ privacy mode: distort facts
15:     $TA_{t+1} \leftarrow M^{\text{TA}}(F_0, F_t, fb_t, I)$  ▷ assign transformation to each fact
16:     $F_{t+1} \leftarrow M^{\text{TR}}(F_t, TA_{t+1}, fb_t, I)$  ▷ apply transformations
17:  else ▷ utility mode: reverse fact transformations
18:     $TA_{t+1} \leftarrow M^{\text{TA}}(F_0, F_t, fb_t, H, I)$ 
19:     $F_{t+1} \leftarrow G(TA_{t+1}, H)$  ▷ apply reversals mechanically
20:  end if
21:   $H.\text{APPEND}([ (TA_{t+1}^i, f_{t+1}^i) : i = 1, \dots, |F_{t+1}| ])$ 
22:   $x_{t+1} \leftarrow M^{\text{DR}}(F_{t+1}, x_0)$  ▷ document rewriting
23: end for
24: return  $\text{BEST\_TEXT}((x_0, p_0, u_0), \dots, (x_T, p_T, u_T), \tau)$ 

```

Any appendices, including any required ethics information, should be included after the references.

Markers do not have to consider appendices. Make sure that your contributions are made clear in the main body of the dissertation (within the page limit).

Appendix B

Participants' information sheet

If you had human participants, include key information that they were given in an appendix, and point to it from the ethics declaration.

Appendix C

Participants' consent form

If you had human participants, include information about how consent was gathered in an appendix, and point to it from the ethics declaration.